

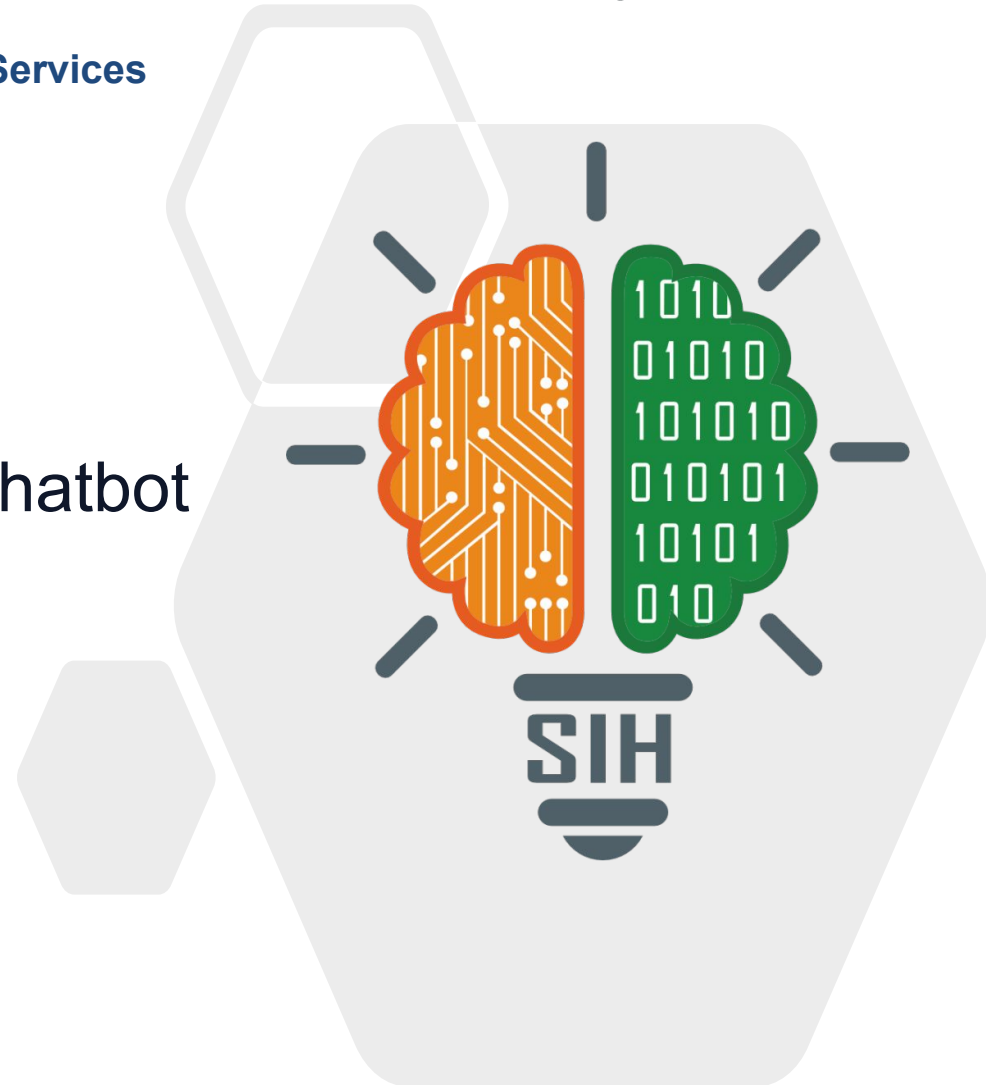
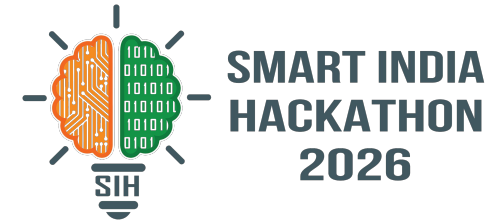
SMART INDIA HACKATHON 2026

CO-OP MITRA

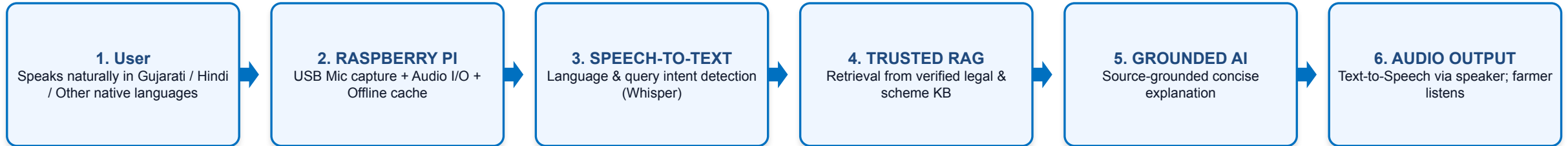
Edge-AI Voice Assistant for Rural Cooperative Services

"Speak. Understand. Access. Empower."

- **Problem Statement ID** – 26088
- **Problem Statement Title** – Multilingual Cooperative Governance & Legal Assistance Chatbot
- **Theme** – Agriculture, FoodTech & Rural Development
- **PS Category** – Hardware
- **Team ID** –
- **Team Name** – LogicSmiths



Proposed Solution



Schemes & PMFBY

Crop insurance eligibility, sowing deadlines, required documents, claim status, and subsidy rules grounded in official guidelines.



Laws & By-Laws

Multi-State & State Cooperative Acts and Model PACS Bye-Laws (2023) translated into multiple native-languages actionable steps.



Financial Literacy

Guidance on PACS credit facilities, KCC interest subvention (3%), Kisan credit limits, transparent interest rates & savings schemes.



Grievance Assistance

Voice-to-draft structured complaint registration prototype for delayed loans, fertilizer allocation, or cooperative service issues.

How It Addresses Rural Barriers (Why Different?):

- **Voice-First Inclusion:** Zero typing or reading needed; bridges rural digital illiteracy.
- **Vernacular Interaction:** Native speech in Gujarati, Hindi & many more languages for natural query flow.
- **Strict Source-Grounding:** Verified legal gazette citations; Hallucination-controlled, source-grounded responses.
- **Offline Local KB Fallback:** Essential FAQs and rules available even during network outages.
- **Physical Edge Access:** Deployed at PACS centers; eliminates intermediary reliance.

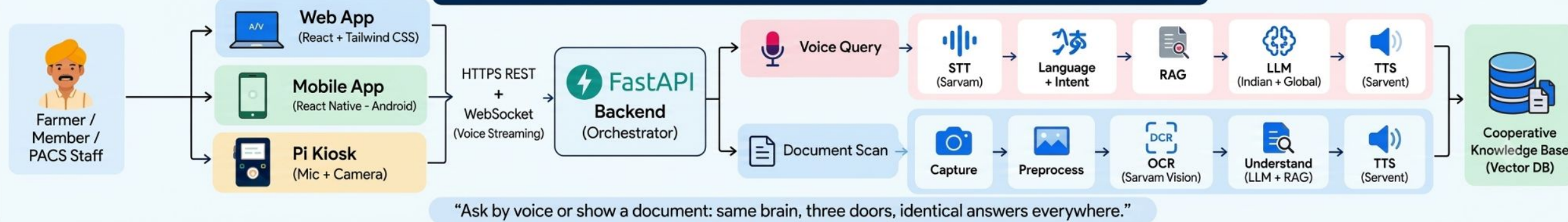


Innovation & Uniqueness of Solution:

"Not just another software chatbot — a low-cost physical AI access terminal that brings cooperative governance and legal rights directly to the rural users."

Combines edge audio hardware (Raspberry Pi + Mic + Speaker) with hybrid cloud AI and offline local storage to ensure uninterrupted service delivery in remote villages.

ONE BACKEND. THREE CHANNELS. VOICE + DOCUMENT SCAN. INDIAN AI.



1 Web App (React + Tailwind CSS)

React (Vite)
Tailwind CSS

- Chat + mic, multilingual
- Upload document for explanation
- Admin panel (grievances, KB updates)

2 Mobile App (React Native - Android)

React Native (Expo)
Android (First)

- Voice-first chat with audio playback
- Snap document and get explanation
- Language picker for scheme, Marathi + more
- Push notifications for scheme deadlines
- Grievance form (auto-filled)

3 Pi Kiosk with Document Scanner

Hardware:

- Raspberry Pi 4 (4GB)
- 1080p Webcam
- LED Light
- USB Mic
- SW Speaker

Software:

- Python 3.11
- OpenCV (Auto capture)
- RNNNoise (Audio)
- WebSocket (Client)

Labels in image: Camera (Arma Camera), Mini Display (Connected to Raspberry Pi), Raspberry Pi (with case), Speaker (for audio output), Microphone (for voice input)

Backend & Indian AI Models

Backend FastAPI Python	STT (Speech → Text) Sarvam STT	LLM (Indian → Global) Sarvam OpenAI Hugging Face	OCR (Documents) Sarvam Vision	TTS (Text → Speech) Sarvam TTS
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Knowledge Base & RAG

Govt. Schemes Guidelines Forms / Notices Official Portals

PGVECTOR (Vector Database)

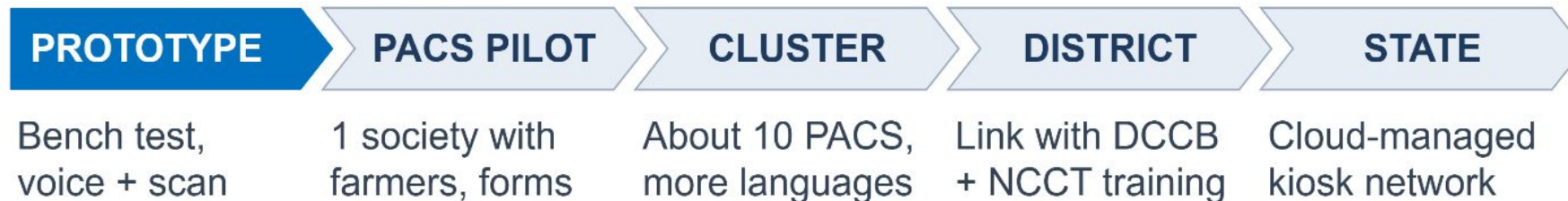
Implementation Methodology

- Requirement Analysis
- System Design & Setup
- Development & integration
- Testing & Pilot
- Deployment & Feedback

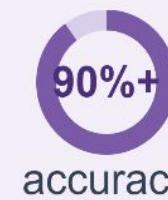
Prototype Link: <https://co-op-ai-mitra.vercel.app>

FEASIBILITY	CHALLENGES / RISKS	MITIGATION
1 Proven, mature stack FastAPI, OpenCV, FAISS RAG	1 Accents hurt speech accuracy	✓ Indian STT + read-back
2 Indian AI available today Sarvam, Bhashini, AI4Bharat	2 OCR errors on poor scans	✓ Blur check + OCR fallback
3 Scan it, hear it Indic OCR + explanation in voice	3 Wrong or misread answers	✓ Strict RAG + read-back
4 Grounded, verified answers RAG on Acts, Bye-Laws, PMFBY	4 Noise at busy PACS centers	✓ Directional mic + RNNoise
5 Low-cost, low-power kiosk Pi 4 (5V/3A), approx. Rs 12-15K	5 Privacy of voice and papers	✓ Auto-delete, mask, DPDP
6 Voice-first, shared use No typing; one kiosk per PACS	6 Weak internet or AI outage	✓ Retry, FAQ cache, fallbacks
	7 Scheme rules keep changing	✓ Admin upload, auto re-index

PHASED ROLLOUT PLAN



PILOT TARGETS (not results)



Voice reply < 5 s
Scan read 10-15 s
Kiosk < Rs 15K

The information exists. **Members just can't understand it in their language, at the moment they need it.**



WHO BENEFITS



FARMERS & PACS MEMBERS

Understand loans, KCC, PMFBY papers and their rights by voice, in their own language.



PACS STAFF & SECRETARIES

Fewer repetitive "what does this paper say?" queries; more time for real service.



COOPERATIVE OFFICIALS & TRAINERS (NCCT)

A scalable awareness tool and a view of what members ask about.

KEY IMPACT



SOCIAL IMPACT



Inclusion: language and literacy stop blocking access to rights.



Any paper can be read aloud and explained, no typing needed.



Members know their rights and how to raise a grievance.



ECONOMIC IMPACT



Fewer missed scheme benefits and insurance deadlines (e.g. PMFBY 72-hour window).



Fewer form mistakes and rejections from misunderstood fields.



Less dependence on middlemen, so fewer informal fees and errors.



GOVERNANCE & TRANSPARENCY



Explanations cite the official Act / bye-law clause, so guidance is consistent everywhere.



Grievances easier to raise and track, so more accountability.



Supports PACS computerization and digital inclusion push.

PILOT SUCCESS METRICS (TARGETS)

Targets, to be measured in PACS pilot



Answer accuracy
90%+
on tested questions



Correct reading of
clean printed forms



Voice response
< 5 seconds



Fewer repeat queries
at PACS counter



Member satisfaction
from pilot feedback

POLICY AND DOMAIN SOURCES



Ministry of Cooperation

PACS Model Bye-Laws & Cooperative Acts



PMFBY

Crop Insurance Guidelines & Grievance Framework



NABARD & NCDC

Cooperative Credit & Financial Literacy Guidelines

TECHNOLOGY SOURCES



Raspberry Pi Documentation

Edge Computing & Audio Architecture



AI/ML Frameworks

Whisper, RAG & Indic-TTS for multilingual assistance



Bhashini

Indian Language AI & Speech Technologies

co-op-ai-mitra.vercel.app



LIVE PROTOTYPE

Try Co-op Mitra: scan a paper, hear it explained

<https://co-op-ai-mitra.vercel.app>



Deployed on Vercel